System Architecture

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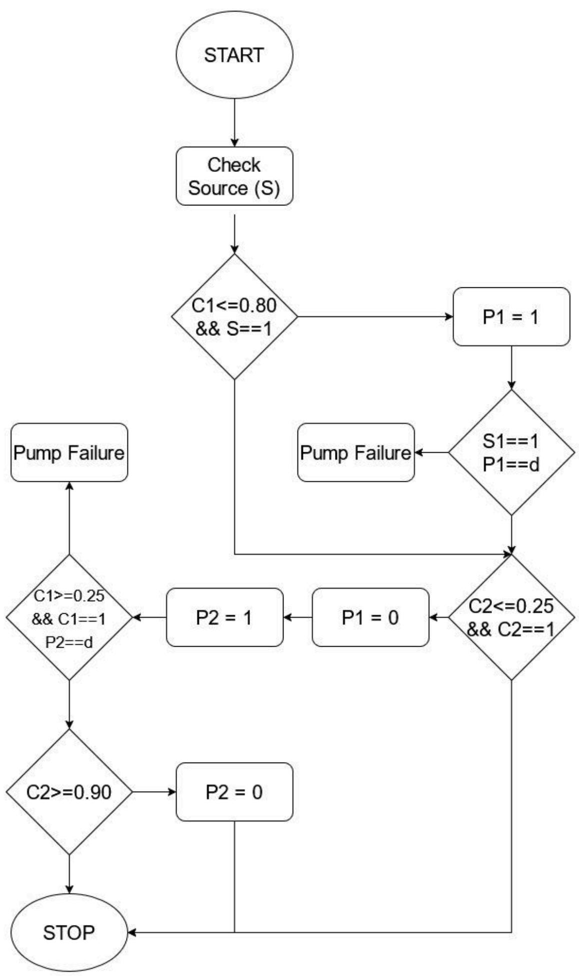
Logic Schematic

Operational Thresholds & Setpoints

To ensure system longevity and prevent critical "stock-out" events, the system operates within defined Operational Envelopes. Rather than utilizing absolute zero (0%), the system maintains a safety buffer to prevent sediment ingestion and pump cavitation.

Storage Unit	Low-Level Setpoint (LL)	High-Level Setpoint (LH)	Purpose
Reservoir (C1)	25%	80%	Buffer for municipal fluctuations.
Overhead Tank (C2)	25%	80%	Buffer for peak domestic demand.

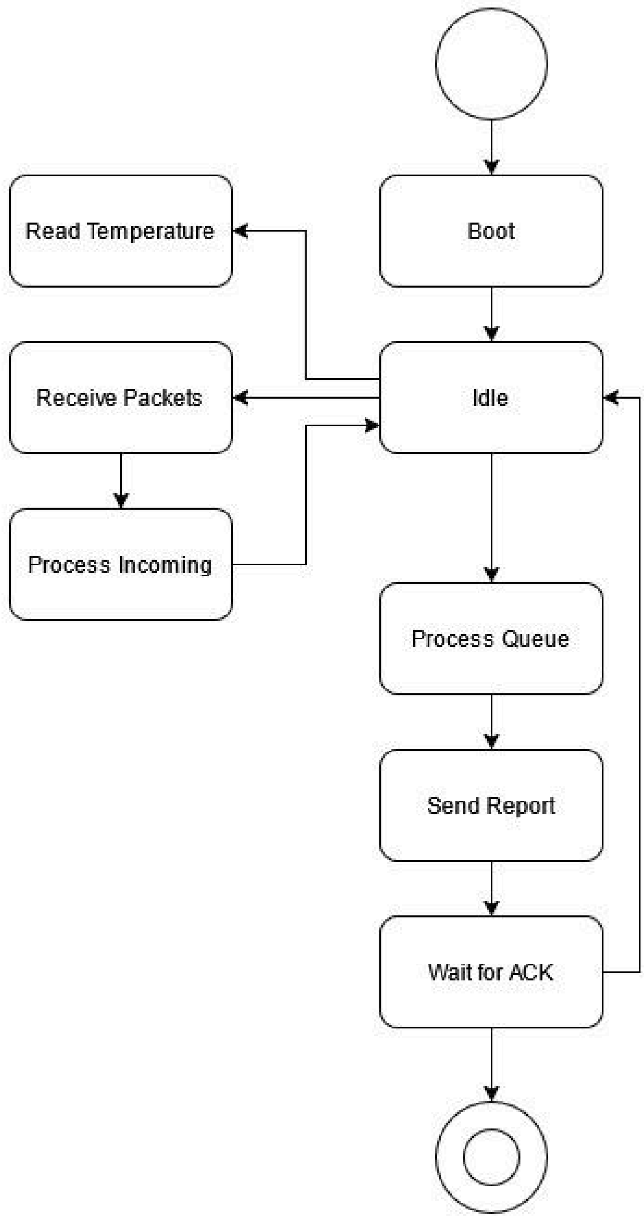


Logic Flow Chart

System States

The logical behavior of the BareBone OS is defined by three distinct Finite State Machines (FSMs) operating in parallel across the network. The following sections detail the operational flow for the Overhead Station (OHS), Sump Station (SS), and the Control Station.

Overhead Station (OS) - The Event Generator



OHS State Transition Diagram

SC1 States

1. BOOT — Initialize pins, begin serial and HC12, begin DHT11, take baseline ultrasonic reading
2. IDLE — Main loop, waiting for triggers
3. READ_TEMPERATURE — DHT11 reading every 10 minutes, only if channel clear
4. READ_LEVEL — JSN-SR04T fires, calculates distance using current temperature, returns percentage
5. RECEIVE_PACKET — 6 bytes available on HC12, read and validate preamble, destination, checksum
6. PROCESS_INCOMING — Route valid packet, handle CMD_PING or CMD_ACK
7. SEND_REPORT — Build and push level report to queue
8. WAIT_FOR_ACK — Wait up to 630ms, retry up to 3 times, discard gracefully on failure

The Overhead Station (OHS) operates on a Wake-on-Event architecture designed to minimize radio traffic and ensure high-precision telemetry. The system logic follows these primary paths:

External Trigger (Polling & Synchronization) When a "PING" or "GET_LEVEL" packet is received from the Master, OHS transitions to the **PACKET RECEIVED** state. It immediately generates a health report. To prevent collisions, OHS utilizes a 1300ms Backoff Timer before attempting transmission, ensuring it does not overlap with SS or Master broadcasts.

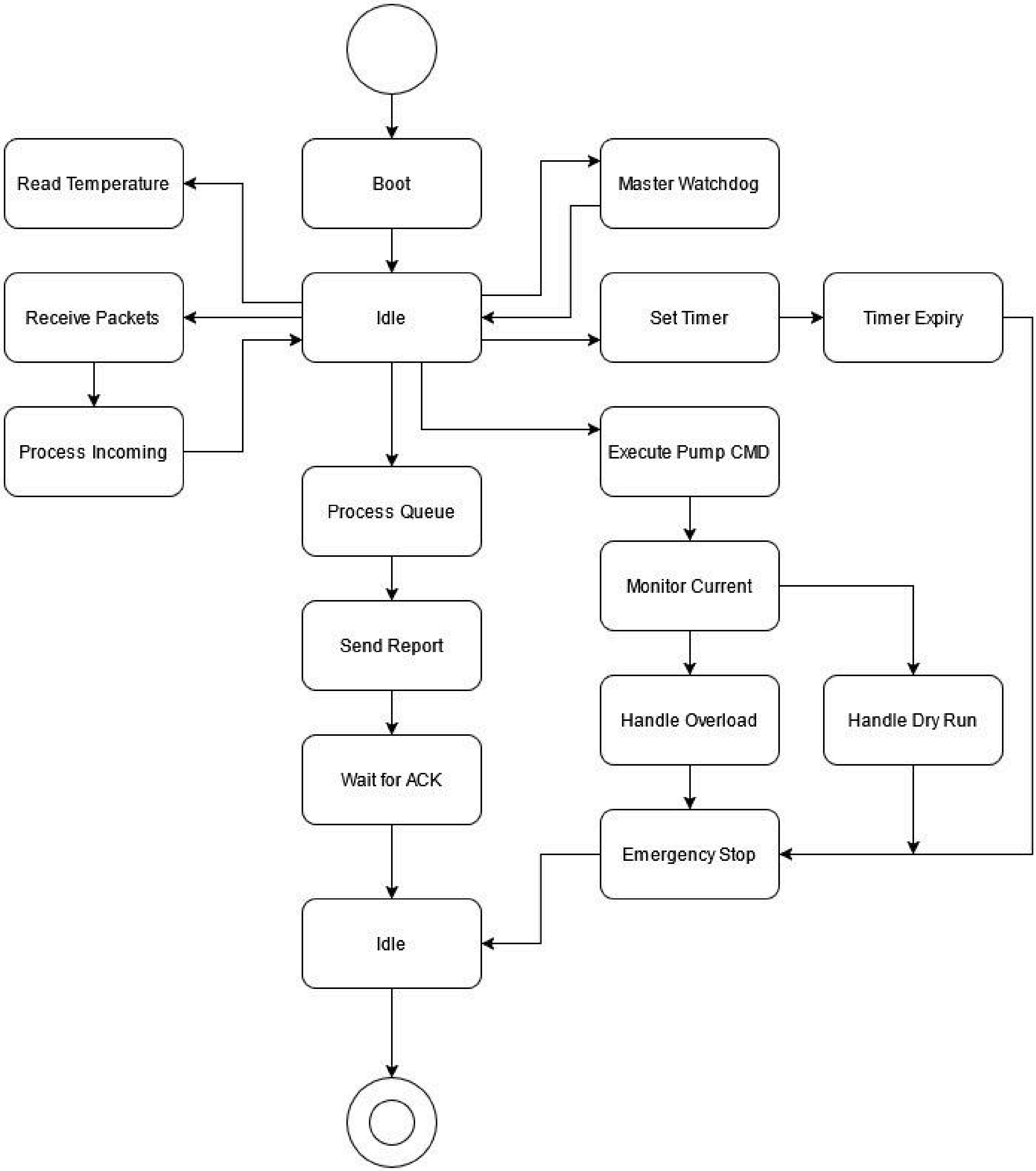
Internal Trigger (Environment & Level Change)

- Precision Level Sensing: OHS utilizes a JSN-SR04T ultrasonic sensor. Every 10 minutes (provided the HC-12 channel is clear), a DHT11 sensor triggers a temperature reading. The speed of sound is dynamically recalculated using the formula: $v=331.3+(0.606\times temp)$. This ensures the 80% TANK_HIGH threshold is accurate regardless of ambient weather conditions.
- Asynchronous Event Logic: If a significant level shift is detected, OHS bypasses standard polling and pushes a report to the 8-slot Priority Queue.

Priority Transmission & Reliability The "Transmission Queue" is now a Priority-Based Stack. Packets are categorized as follows:

- Priority 1/2: Critical Faults or State Changes.
- Priority 3: Normal Level Reports.
- Priority 4: ACKs. The WAITING FOR ACK state remains the final safeguard, confirming the Master has logged the data before OHS clears that specific slot from the queue.

Sump Station (SS) - The WorkHorse



SS Transition Diagram

SC2 States

1. **BOOT** — Initialize pins, relays OFF, begin serial and HC12, begin DHT11, take baseline ultrasonic reading
2. **IDLE** — Main loop, waiting for triggers
3. **READ_TEMPERATURE** — DHT11 reading every 10 minutes, only if channel clear
4. **READ_LEVEL** — JSN-SR04T fires, calculates distance using current temperature, returns percentage
5. **RECEIVE_PACKET** — 6 bytes available on HC12, read and validate preamble, destination, checksum
6. **PROCESS_INCOMING** — Route valid packet, handle CMD_PING, CMD_PUMP, CMD_TIMER, CMD_ACK
7. **EXECUTE_PUMP_COMMAND** — Activate or deactivate relay based on command value, respect locked and fault flags
8. **SET_TIMER** — Store timer end timestamp, set timerActive flag
9. **TIMER_EXPIRY** — Timer end reached, P2 OFF, timerActive cleared, report pump state
10. **MONITOR_CURRENT** — Read ACS712 via multiplexer for active pump, check against dry run and overload thresholds
11. **HANDLE_DRY_RUN** — Pump OFF, increment retry count, wait 10 seconds, retry up to 3 times before escalating
12. **HANDLE_OVERLOAD** — Both pumps OFF immediately, no retries, escalate to emergency stop
13. **EMERGENCY_STOP** — Cancel timer, kill both relays, push CMD_FAULT and CMD_AMP to queue at critical priority
14. **SEND_REPORT** — Build and push level report to queue
15. **PROCESS_QUEUE** — Transmit highest priority packet, handle retries on 500ms backoff
16. **MASTER_WATCHDOG** — Monitor time since last Master packet, freeze P2 after 2 minutes silence, report on reconnect

The Sump Station (SS) acts as the system's "Intelligence & Power Hub," managing high-power actuators and multi-channel safety sensors. The architecture is governed by an advanced feedback and protection loop:

To ensure industrial-grade current monitoring, the Sump Station (SS) utilizes an ADS1115 16-bit external ADC. This upgrade replaces the internal 10-bit/12-bit non-linear ADCs typically found in microcontrollers, providing a significantly higher dynamic range for the dual ACS712 current sensors.

- Dual-Channel Monitoring: The system independently monitors Pump 1 (P1) and Pump 2 (P2) through dedicated differential channels on the ADS1115.
- Elimination of Switching Latency: By removing the previous D7 multiplexing logic, the system can now perform near-simultaneous sampling of both power rails. This allows the RTOS to execute high-speed fault detection algorithms—such as detecting a "locked rotor" or "dry-run" condition—without the signal degradation or propagation delays associated with manual pin-switching.

To maintain the integrity of the 80% TANK_HIGH critical threshold, the Overhead Station (OS) implements an Acoustic Velocity Compensation algorithm.

- Atmospheric Feedback: A DHT11 thermistor/hygrometer samples the ambient temperature and humidity within the tank enclosure every 10 minutes.
- Ultrasonic Calibration: Since the speed of sound varies with air density, the OS uses this atmospheric data to dynamically recalibrate the JSN-SR04T ultrasonic transducer. This ensures that the distance-to-water-level calculation remains accurate to within ±1% despite seasonal temperature fluctuations, preventing premature pump shut-off or overflow due to acoustic drift.

The "Double-Tap" Fault Protocol The SENSOR MONITORING block now performs a more sophisticated analysis. If a Dry Run or Overload is detected:

- Packet 1 (CMD_FAULT): Immediate broadcast of the fault type.
- Packet 2 (CMD_AMP): Follow-up packet containing the exact amperage (Encoded as VAL × 10).
- Priority: These are pushed to Priority 1 (CRITICAL) in the 8-slot queue, bypassing all level reports.

The "Locked" Safety State The system no longer just enters "Emergency Stop." It now implements a `isLocked` flag.

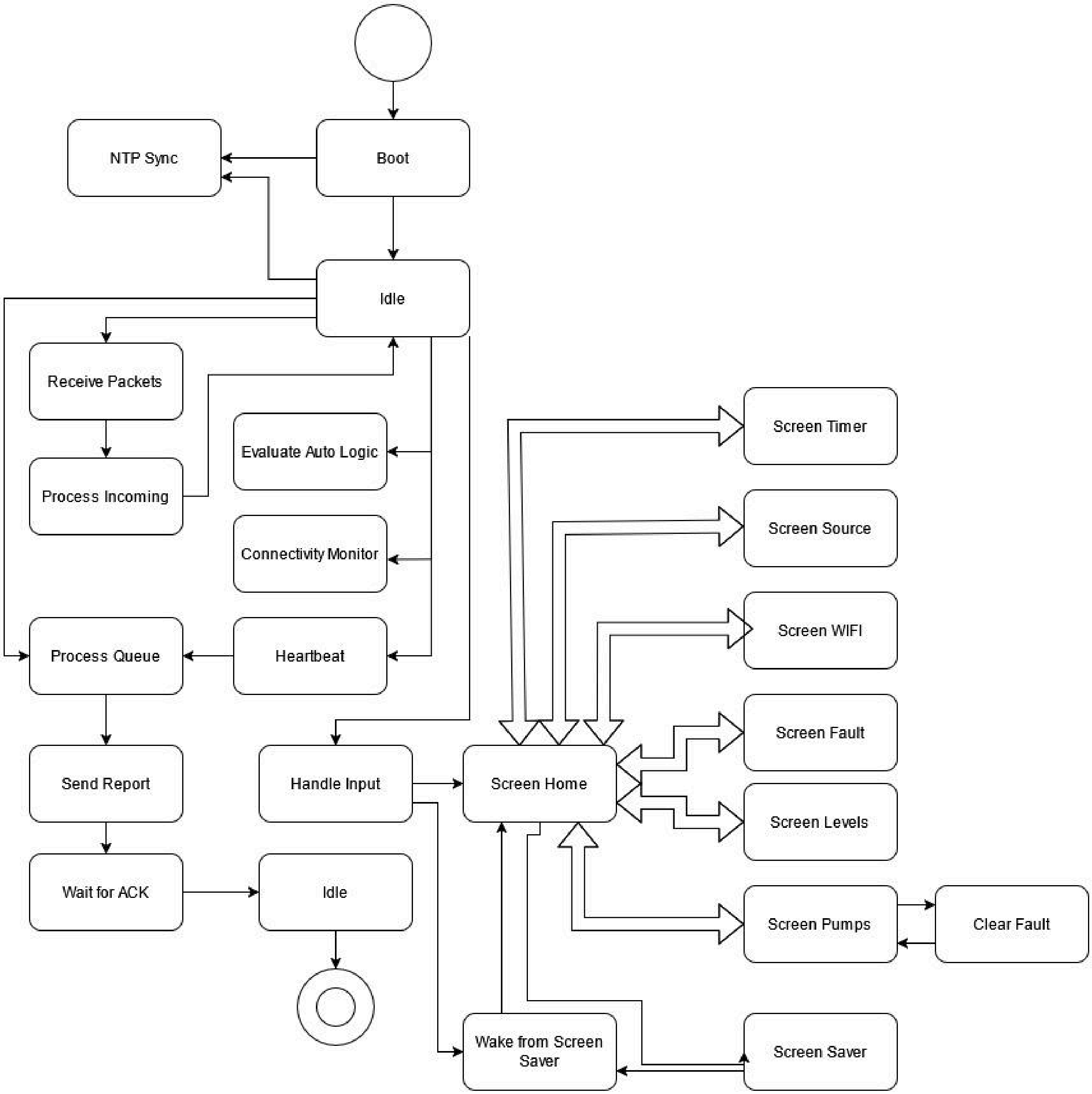
- The Logic: Once a fault occurs, the pump is logically locked. Even if the fault condition clears, the Auto-Logic cannot restart the pump.
- Reset: A manual intervention from the Master’s pump screen is the only way to clear the `isLocked` state.

Communication Watchdog (Fail-Safe) SS monitors a DISCONNECT_PERIOD (120,000ms). If the Master is silent for 2 minutes, the Disconnect Watchdog triggers:

- Action: Pump 2 (P2) is immediately frozen to prevent uncontrolled operation during a comms blackout.
- Recovery: Upon reconnection, SS automatically synchronizes its current P2 state back to the Master.

Collision Management To ensure the Master stays in control, SS uses a 383ms Backoff Timer. This allows it to transmit faster than OHS (521ms) but slower than the Master (250ms), prioritizing reservoir/pump data over overhead tank data.

Master Control Unit - The Central Arbiter



Control Station Transition Diagram

Control States

1. **BOOT** — Initialize pins, begin HC12, init display, draw static frame
2. **NTP_SYNC** — Connect WiFi, get time, disconnect, schedule next sync at 7AM
3. **IDLE** — Main loop, all monitors running simultaneously
4. **SCREENSAVER** — Activated after 60 seconds inactivity, display screensaver images, any button wakes
5. **WAKE_FROM_SCREENSAVER** — Restore static frame, redraw previous screen
6. **UPDATE_CLOCK** — Redraw time string only if time has changed, called every loop
7. **HEARTBEAT** — Send ping to OHS at 1 second mark, SS at 31 second mark, reset on 31 second cycle
8. **RECEIVE_PACKET** — 6 bytes available on HC12, validate preamble, destination, checksum
9. **PROCESS_INCOMING** — Route valid packet, update system state, update display
10. **HANDLE_INPUT** — Read analog A0, route to Toggle, Select or Back handler
11. **SCREEN_HOME** — Menu navigation, cursor cycles through 5 items on toggle
12. **SCREEN_PUMPS** — Pump selection and ON/OFF action, heart cursor, lightning indicators
13. **SCREEN_LEVELS** — Static tank images, live C1 and C2 percentage display
14. **SCREEN_TIMER** — Preset selection, countdown display, cross shown when timer active
15. **SCREEN_SOURCE** — YES/NO source toggle, disabled when timer active or C2 full
16. **SCREEN_WIFI** — SSID and IP display, connection diagnostic with hourglass animation
17. **SCREEN_FAULT** — Fault display with pump name, reason and amperage, buzzer alarm, locked until SELECT
18. **EVALUATE_AUTO_LOGIC** — P1 logic driven by source flag and C1 level, P2 logic driven by C2 demand and C1 availability, respects locked flags
19. **CONNECTIVITY_MONITOR** — 180 second timeout for OHS and SS, kill pumps if SS goes silent
20. **PROCESS_QUEUE** — Transmit highest priority packet, handle retries on 250ms backoff
21. **CLEAR_FAULT** — Lock faulted pump, clear fault code, restore previous screen

The Master Unit acts as the High-Speed System Coordinator and HMI (Human-Machine Interface). It manages the flow of the HydroLogix OS protocol while maintaining a strict safety barrier between user commands and hardware execution.

Instant-Access User Interface (Manual Input) Unlike the satellite nodes, the Master has Zero Boot Delay. The UI is available immediately upon power-up.

- Extended Scheduling: The timer system has been expanded from 3 presets to 9 steps (30-minute increments up to 4 hours 30 minutes), providing finer control over tank filling cycles.
- The Safety Gate: If a pump is in a Locked (`isLocked = true`) state due to a previous fault, the Master will ignore all "Auto" logic requests. The user MUST navigate to the "Pumps Screen" and manually trigger to reset the flag.

Priority-Driven Network Orchestration (Listen Radio) The Master utilizes a high-speed 9600bps baud rate and the most aggressive timing in the system with a 250ms Backoff Timer. Communication is managed via an 8-slot Priority Queue:

1. Priority 1: `CMD_PUMP` (Instant action)
2. Priority 2: `CMD_TIMER` (Scheduling)
3. Priority 3: `CMD_PING` (Health checks)
4. Priority 4: `CMD_ACK` (Confirmations)

Advanced Telemetry & Fault Processing The Master now handles a "Double-Tap" data flow from SC2:

- Packet Reception: When a fault occurs, the Master expects two consecutive packets: CMD_FAULT followed by CMD_AMP_Px.
- Data Decoding: The Master receives the amperage as an integer (VAL × 10) to save bandwidth and decodes it back to a float (VAL / 10.0) for the UI display.
- ACK Logic: The Master holds the Acknowledge (ACK) signal until both the Fault and the specific Amperage data are successfully logged into the local registers.

Heartbeat & Watchdog Maintenance The HEARTBEAT state ensures the Disconnect Watchdog on SC2 remains satisfied. By maintaining a constant polling cycle, the Master prevents SC2 from "freezing" P2. If the Master fails to send a signal for 120 seconds, it assumes a system-wide comms failure and prepares for a re-sync upon reconnection.

Control Algorithm & Logic Flow

The system employs a Calibrated Asynchronous Monitoring Loop. Automation is governed by a safety-first hierarchy where hardware protection (Amperage/Watchdog) overrides all operational commands.

Sensor Calibration Layer (The Foundation)

Before any logic execution, the system validates environment data:

- Temperature Compensation: OHS and SS read the DHT11 every 10 minutes to update the speed-of-sound constant. This ensures the 80% TANK_HIGH threshold is physically accurate.
- Current Sensing Prep: SS toggles Pin D7 to select the correct ACS712 channel (LOW for P1, HIGH for P2) before sampling amperage.

Primary Logic (Municipal Supply → C1)

The system uses the expanded 9-step Timer (30m to 4.5h) or a manual HMI trigger to initiate P1:

- Safety Gate: `IF (P1_isLocked == TRUE) → REJECT START`.
- Execution: Activate P1 + Set D7 LOW.
- Fault Logic: If P1 Amperage (decoded as VAL / 10.0) indicates a Dry Run or Overload, SC2 sends a Double-Tap report (CMD_FAULT + CMD_AMP), sets `P1_isLocked = TRUE`, and shuts down.

Secondary Logic (C1 → C2 Transfer)

The transfer logic follows a Watchdog-Protected Interlock:

- Automation Logic: * START P2 IF: `Level(C2) < 25%` AND `Level(C1) > 25%` AND `P2_isLocked == FALSE`.
    - STOP P2 IF: `Level(C2) ≥ 80%` OR `Level(C1) ≤ 20%`.
- The Watchdog Interrupt: If the 120,000ms Disconnect Watchdog expires (no signal from Master), P2 is immediately frozen in its last safe state (usually OFF) to prevent reservoir depletion during a communication blackout.

Fault Recovery & Priority

- Manual Override: Once a pump enters the `isLocked` state, it cannot be restarted by the timer or auto-logic. A Manual Unlock Command must be sent from the Master's Pump Screen.
- Traffic Priority: Any change in pump state or fault detection generates a Priority 1 or 2 packet, which jumps to the front of the 8-slot queue, ensuring the Master reacts within 250ms of the event.

Semi-Automated Input & Feedback Loop

Due to the non-deterministic nature of municipal water schedules, the system utilizes a High-Priority Manual Interface on the Master Unit. This allows the operator to synchronize automation with supply availability while the system provides real-time safety telemetry.

Synchronized "Supply Windows" The operator no longer simply toggles a switch. Instead, they select from 9 Timer Presets (ranging from 30 minutes to 4 hours 30 minutes in 30-minute steps). This allows the user to match the pumping cycle to the known duration of the municipal supply.

- Immediate Availability: Because the Master Unit has no boot delay, the operator can initiate a cycle the moment municipal water arrives, even if the satellite stations (OHS/SS) are still finishing their 60-second stabilization.

The Safety Feedback Loop (Verification) Once a "Supply Window" is initiated, the system performs an active verification of the operator's input:

- Active Monitoring: As P1 begins drawing from the municipal line, the SS station monitors the ACS712.
- Dry-Line Detection: If the operator acknowledges supply but the line is actually dry, the system detects a low-amperage fault within seconds.
- Lock-Out Response: The system immediately sets the isLocked flag and sends the "Double-Tap" fault

report to the Master. The HMI then shifts from "Pumping" to a "FAULT: P1 LOCKED" state.

Manual Intervention & Reset The feedback loop requires a "Human-Clear" for high-stakes errors:

- Non-Auto Recovery: If a fault occurs during a manual cycle, the timer is cleared, and the pump is locked. The auto-logic cannot "retry."
- Manual Unlock: The operator must manually interact with the Master's pump screen to Unlock the system. This ensures the operator has physically checked the municipal line or pump intake before the system allows another attempt.

Command Priority Every manual input (Start, Stop, Timer Change, Unlock) is pushed into the Priority 1 (CRITICAL) slot of the Master's 8-slot queue. This ensures that the user's command is transmitted within 250ms, providing the "snappy" feel of a hard-wired switch despite the wireless architecture.

Fault Diagnostics & Error Handling

The system utilizes an Advanced Telemetry & Safety Layer designed to protect expensive pump hardware and prevent water wastage. Diagnostics are no longer just "binary" (on/off); they are now quantitative, providing real-time amperage data for every failure.

Expanded Failure Modes (The "Six-Point" Check)

The system differentiates between specific electrical and hydraulic failures using the ACS712 sensors:

- Dry Run (P1_DRY / P2_DRY): Detected when Amperage < *Lower Threshold*. Indicates the municipal line is dry or the reservoir (C1) is empty.
- Overload (P1_OVER / P2_OVER): Detected when Amperage > *Upper Threshold*. Indicates a jammed impeller, internal short, or motor fatigue.
- Hardware/Open Circuit (P1_FAIL / P2_FAIL): Detected when the relay is triggered but Amperage is 0.0A. Indicates a blown fuse, tripped breaker, or broken wiring.

The "Double-Tap" Reporting Protocol

To save bandwidth while providing maximum detail, the system uses a two-packet high-priority burst:

1. Packet A (CMD_FAULT): Immediate notification of the fault type (Priority 1).
2. Packet B (CMD_AMP_Px): Follow-up packet containing the exact current reading at the moment of failure (Encoded as VAL × 10). The Master unit decodes this (VAL / 10.0) and displays the exact trip-current on the screen (e.g., *"FAULT: P1 OVERLOAD @ 8.4A"*).

The "Locked" Safety State

Every fault listed above triggers the `isLocked` flag for the affected pump.

- Logical Isolation: Once a pump is locked, the Master removes it from the automation pool.
- Manual Clearance: To prevent "fault-looping" (where a system keeps trying to restart into a short circuit), the error must be manually acknowledged and UNLOCKED from the HMI.

Communication Watchdog (System Vitality)

A new "Silent-Failure" diagnostic has been added:

- DISCONNECT_PERIOD (90s): If the Master fails to "Ping" the system for 2 minutes, SS identifies a Comms Timeout.
- Fail-Safe Action: SS freezes Pump 2 (P2) to prevent the reservoir from being drained into the roof while the system is "blind."
- Auto-Recovery: Once the Master reconnects, the priority queue synchronizes the current pump states to ensure the UI matches the hardware.

Array Holders

The system employs several structured arrays across its stations to manage state, communication, and user interface data in a clean and maintainable manner. Rather than relying on scattered individual variables, related data is grouped into arrays and structures to allow indexed access, reduce code duplication, and improve overall readability. These arrays are declared globally and persist throughout the lifetime of the program, serving as the single source of truth for their respective domains.

Pump State Array The **pumps[ ]** array, present on both the Control and SS stations, is a two-element array of the PumpState structure. Each element corresponds to one pump — index 0 for the Overhead pump and index 1 for the Basement pump. The structure encapsulates all relevant per-pump information including its current on/off state, fault status, lockout flag, fault code, dry run retry count, and a human-readable status label. Centralizing pump data in this manner ensures that both the auto logic and the display functions always reference a single consistent source.

Packet Queue Array The **packetQueue[ ]** array, present on both the Control and SS stations, is an 8-slot array of the QueuedPacket structure. Each entry stores the raw 6-byte packet data alongside its assigned priority level, current retry count, and acknowledgement status. This array forms the backbone of the priority-based transmission system, ensuring that critical fault packets are always dispatched ahead of routine reports and acknowledgements.

Timer Preset Arrays The **timerPresets[ ]** and **timerLabels[ ]** arrays are present exclusively on the Control station and together define the nine available timer presets. The **timerPresets[ ]** array stores the duration of each preset in minutes ranging from 30 to 270 in 30-minute increments, while **timerLabels[ ]** stores the corresponding human-readable strings displayed on the timer screen such as "1 hr", "2 hr 30 min" and so on. Both arrays are indexed together using a single timerIndex variable that advances on each toggle press.

Packet Buffer Arrays The **inPacket[ ]** and **outPacket[ ]** arrays are present on all three stations and each hold exactly 6 bytes corresponding to the fixed WACP v4.0 packet structure. The **outPacket[ ]** buffer is populated by the packet builder before each transmission, while i**nPacket[ ]** receives and holds incoming bytes pending preamble validation, destination verification, and checksum confirmation. These buffers are intentionally kept minimal at 6 bytes each to reduce RAM overhead on the memory-constrained NodeMCU platform

System Labeling

Microcontroller Unit (MCU): NodeMCU (ESP8266) & ESP32

The system architecture is built upon a distributed heterogeneous processing model, strategically employing both the ESP32 and ESP8266 (NodeMCU) platforms to balance high-performance edge computing with cost-effective telemetry. The core of the system—comprising the Control Station and the Sump Station (SS)—has been upgraded to the dual-core ESP32 (Xtensa® 32-bit LX6) to facilitate a transition from simple sequential execution to a Preemptive Multitasking environment via FreeRTOS. This dual-core capability allows for "Task Splitting," where time-critical safety operations, such as high-speed current sampling through the ADS1115, run on a dedicated core to ensure sub-millisecond fault detection, while the HMI and system orchestration tasks remain isolated on the second core to prevent UI latency. Conversely, the Overhead Station (OS) utilizes the ESP8266 as a dedicated Edge Sensor Node; its high performance-to-cost ratio and integrated Wi-Fi stack make it ideal for processing acoustic telemetry and environmental calibration data before transmitting it to the orchestrator.

While industrial Programmable Logic Controllers (PLCs) were considered for their noise immunity, they were ultimately rejected due to their prohibitive unit costs and rigid hardware overhead which would limit the custom algorithmic control required for features like Acoustic Velocity Compensation. Instead, the design achieves "industrial-parity" through rigorous hardware engineering, including custom-designed PCBs featuring EMF shielding and AC/DC galvanic isolation to maximize creepage and clearance distances. By selecting ESP-class controllers, the project maintains the rapid iteration cycles of embedded development while ensuring the control logic is structured for potential future migration to IEC 61131-3 PLC environments. This hybrid approach ensures that the system remains scalable and cost-efficient without compromising the high-fidelity signal acquisition and fault-tolerance required for critical domestic infrastructure.

Power Distribution: Isolated SMPS & Protection

The system's power architecture is engineered for high-reliability performance, utilizing a multi-stage regulation and protection strategy to ensure both user safety and logic-level integrity. Primary power conversion is handled by an Isolated Switch-Mode Power Supply (SMPS), which steps down the 220V AC utility voltage to a regulated 5V DC rail. This isolation stage acts as the first line of defense, physically decoupling the high-voltage municipal supply from the sensitive low-voltage control circuitry.

To ensure a stable "clean" power rail free from the transients typical of inductive motor switching, the PCB incorporates a robust distributed decoupling network:

- Bulk Capacitance Reservoir: Dual 220µF electrolytic capacitors are strategically placed at the 5V SMPS output stage. These serve as a local energy buffer, providing the necessary current "headroom" to absorb voltage sags caused by relay actuation and preventing brown-out conditions during peak load transitions.
- Localized High-Frequency Filtering: For the MCU logic, 10µF decoupling capacitors are situated in immediate proximity to the Vin pins of the ESP32 and ESP8266 processors. These capacitors mitigate high-frequency noise and switching ripple, ensuring that the VCC rail remains within the tight tolerances required for stable RTOS operation and accurate ADC sampling.
- Creepage and Clearance Optimization: The high-voltage AC traces are strictly segregated from the DC logic planes through the implementation of milled isolation slots in the PCB substrate. This effectively increases the creepage distance beyond the standard limits of FR4 material to prevent surface tracking and high-voltage arcing.
- EMI Mitigation: Given the proximity to high-power inductive loads (the water pumps), the PCB features strategic EMF shielding. This includes a dedicated ground plane architecture to ensure that electromagnetic noise from the pump motors does not propagate back into the high-precision analog signal path of the ADS1115.

Level Sensing: JSN-SR04T Waterproof Transducer

Unlike mechanical float switches, which are prone to fouling and corrosion, the JSN-SR04T uses non-contact ultrasonic ranging. The IP67-rated waterproof probe is essential for high-humidity environments inside the reservoir, ensuring reliability against condensation and water splashes.

Diagnostic Sensing: ACS712-30A Hall-Effect Current Sensor

A current-based diagnostic approach was chosen over mechanical flow sensors to minimize plumbing modifications and reduce costs. By monitoring the current signature of the pump, the system can detect:

- Under-load: Indicating dry-running (low current).
- Over-load: Indicating mechanical blockage or stall (high current).

Communication: HC - 12433 SI4463

To overcome the high signal attenuation inherent in reinforced concrete structures and multi-floor residential environments, the system utilizes a Sub-GHz wireless mesh-point topology powered by the HC-12 (SI4463-based) transceiver. Operating at the 433.4 MHz ISM band, this module provides superior diffraction and penetration capabilities compared to 2.4 GHz solutions (Wi-Fi/Bluetooth), ensuring link stability through structural barriers.

The communication backbone utilizes Gaussian Frequency Shift Keying (GFSK), which offers high noise immunity and spectral efficiency. The transceivers are configured with a High-Sensitivity Receiving Mode (-112 dBm), allowing the system to maintain a reliable link budget even when the Sump Station and Overhead Station are separated by multiple slabs of high-density concrete.

The interface between the MCUs and the RF modules is managed via a UART-to-Radio Bridge.

- Baud Rate Strategy: The system is standardized at a 9600 bps baud rate. This specific rate was selected to balance data throughput with receiver sensitivity; lower baud rates increase the "Energy per Bit," significantly reducing the Bit Error Rate (BER) over long distances.
- Buffer Management: Within the RTOS environment of the ESP32, the communication task utilizes a dedicated circular buffer to handle incoming packets. This ensures that telemetry from the Overhead Station is parsed asynchronously without interrupting the real-time current monitoring of the pumps.

The modules utilize a TCXO (Temperature Compensated Crystal Oscillator) base within the SI4463 silicon, ensuring that the carrier frequency does not drift due to the high-temperature fluctuations often found in outdoor overhead tank enclosures or humid basement environments.

Actuation Module: 30A Opto-Isolated Relay Module

Domestic water pumps have a high inductive inrush current (initial surge) that can weld the contacts of standard 10A hobbyist relays. A 30A-rated relay provides a necessary factor of safety. The optocoupler ensures galvanic isolation, preventing high-voltage spikes from back-feeding into the NodeMCU's sensitive logic.

Signal Conditioning: (Resistor Voltage Dividers)

The NodeMCU operates on 3.3V CMOS logic, whereas the JSN-SR04T and ACS712 sensors output 5V TTL signals. Integrating level shifters prevents over-voltage stress on the ESP8266 GPIO pins, ensuring the long-term reliability of the silicon.

Temperature & Humidity Module: DHT-11 Sensor

Since ultrasonic distance measurement depends on the speed of sound in air, which varies with environmental temperature and humidity, a temperature–humidity sensing module was integrated to compensate for these variations. By dynamically adjusting the speed of sound used in distance calculations, measurement accuracy and system reliability are improved under changing atmospheric conditions.

$v \approx 331.3 + (0.606 \times T)$

Human-Machine Interface (HMI) Module

To facilitate real-time monitoring and manual intervention, a centralized HMI node is implemented with the following peripherals:

Human-Machine Interface (HMI): 2.4-inch Intelligent Graphical Display

The system’s primary visual interface has been upgraded from a legacy character-based LCD to a 2.4-inch Thin-Film Transistor (TFT) Liquid Crystal Display. This transition facilitates a high-fidelity Graphical User Interface (GUI), allowing the "Pocket OS" to move beyond simple text strings into comprehensive data visualization.

Unlike 20x4 character displays that are limited by fixed-width rows, the 2.4-inch TFT provides a high-resolution canvas for a multi-layered information hierarchy. This allows the simultaneous rendering of:

- Real-Time Graphical Level Indicators: Dynamic bar graphs representing the volumetric state of both the Overhead and Sump stations.
- High-Frequency Diagnostics: Real-time current (Ampere) readouts and "Heartbeat" icons indicating the status of the Sub-GHz wireless link.
- State-Machine Annunciation: Visual cues for system modes (Auto, Manual, Maintenance) and clear, color-coded fault alerts (e.g., "DRY RUN" or "OVERLOAD").

The display serves as the visual engine for a custom-built Embedded OS, managed via a tactile tri-button interface. This setup enables a nested menu system where users can calibrate sensor offsets, set custom 80% TANK_HIGH thresholds, and view historical fault logs without needing an external computer. The inclusion of a frequency-modulated buzzer provides auditory feedback for every UI interaction, creating a professional "Active Response" environment that confirms user inputs in real-time.

The display utilizes a High-Speed Serial Peripheral Interface (SPI), significantly reducing the bus contention compared to standard I2C character displays. This allows the Control Station's RTOS to push frame updates at high refresh rates, ensuring that the HMI remains responsive even while the background tasks are performing intensive floating-point calculations for the Acoustic Velocity Compensation.